

**CS23532**

# **COMPUTER NETWORKS – LAB**

## **PRACTICAL – 2**

### **STUDY OF DIFFERENT TYPES OF NETWORK CABLES**

#### **AIM**

To study different types of network cables, understand their construction and characteristics, compare their advantages and limitations, and learn how to make and test Ethernet straight-through and crossover cables.

## **1. Introduction**

A network cable is a physical communication medium used to carry data between computers, switches, routers, access points and other network devices. Different cables are designed for different transmission speeds, distances, environments and applications.

The main cable families studied in this practical are twisted-pair cable, coaxial cable and fibre-optic cable. Additional modern cable types such as DAC and AOC are also included for broader understanding.

### **1.1 Objectives**

- Identify the major types of network cables used in computer networks.
- Understand the construction and working principle of UTP, STP, coaxial and fibre-optic cables.
- Compare cable categories according to speed, distance, cost and applications.
- Understand the difference between straight-through and crossover Ethernet cables.
- Learn the T568A and T568B colour codes.
- Understand the procedure for terminating an Ethernet cable with an RJ45 plug.
- Test the completed cable and identify common wiring faults.

### **1.2 Major Types of Network Cables**

| Cable Type                  | Signal     | Main Feature                           | Typical Application                 |
|-----------------------------|------------|----------------------------------------|-------------------------------------|
| UTP                         | Electrical | Low cost and easy installation         | Ethernet LANs                       |
| STP / Shielded twisted pair | Electrical | Better protection from interference    | Industrial / high-interference LANs |
| Coaxial                     | Electrical | Shielded single-conductor construction | TV, broadband, CCTV, RF             |

| Cable Type   | Signal     | Main Feature                        | Typical Application               |
|--------------|------------|-------------------------------------|-----------------------------------|
| Fibre optic  | Light      | High bandwidth and EMI immunity     | Backbone, data centres, telecom   |
| DAC / Twinax | Electrical | Short high-speed interconnect       | Data centres                      |
| AOC          | Optical    | Pre-terminated optical interconnect | Data centres and high-speed links |

## 2. Twisted-Pair Cable

Twisted-pair cable consists of insulated copper conductors arranged in pairs. The conductors are twisted around each other to reduce electromagnetic interference and crosstalk. Ethernet cabling normally contains four twisted pairs.

### 2.1 Unshielded Twisted Pair (UTP)

UTP has no metallic shield around the individual pairs or around the complete cable. It is widely used because it is inexpensive, flexible and easy to install.

#### Advantages

- Low cost.
- Easy to install and terminate.
- Flexible and lightweight.
- Widely supported by Ethernet equipment.
- Suitable for many office and laboratory LAN installations.

#### Disadvantages

- More affected by electromagnetic interference than shielded cable.
- Cable performance depends strongly on correct installation and termination.
- Copper Ethernet has a limited channel distance.

### 2.2 Shielded Twisted Pair (STP)

STP uses metallic shielding to reduce the effect of external interference. Shielding can be applied around individual pairs, around groups of pairs, or around the complete cable depending on the cable construction.

- Better resistance to interference and noise.
- Useful in electrically noisy environments.
- Generally more expensive than UTP.
- Requires greater care during installation and grounding.

### 2.3 Other Shielding Forms

| Type         | Meaning / Construction                                   | Main Benefit                                   |
|--------------|----------------------------------------------------------|------------------------------------------------|
| F/UTP or FTP | Overall foil shield around the twisted pairs             | Improved protection from external interference |
| S/UTP        | Overall braided shield with unshielded pairs             | Additional overall shielding                   |
| S/FTP        | Overall braided shield plus foil around individual pairs | Strong shielding and pair isolation            |

## 3. Ethernet Cable Categories

Cable category indicates the electrical performance of the cable. The actual network speed also depends on the Ethernet standard, cable length, connectors, installation quality and network devices.

| Category | Common Speed / Use                               | Typical Reach                   | Notes                                              |
|----------|--------------------------------------------------|---------------------------------|----------------------------------------------------|
| Cat3     | 10 Mbps / 10BASE-T                               | Up to 100 m                     | Older Ethernet category                            |
| Cat5     | 100 Mbps; older installations may support 1 Gbps | Up to 100 m                     | Largely replaced by Cat5e                          |
| Cat5e    | 1 Gbps; supports some 2.5/5 Gbps systems         | Up to 100 m                     | Very common for Gigabit Ethernet                   |
| Cat6     | 1 Gbps; 10 Gbps for shorter links                | Up to 100 m depending on speed  | Better performance margin than Cat5e               |
| Cat6A    | 10 Gbps                                          | Up to 100 m                     | Common choice for 10GBASE-T                        |
| Cat7     | 10 Gbps class                                    | Up to 100 m in suitable systems | Usually shielded; higher-performance cabling class |
| Cat8     | 25/40 Gbps class                                 | Typically up to 30 m            | Mainly high-speed data-centre copper               |

## 4. Coaxial Cable

Coaxial cable has a central copper conductor, an insulating dielectric, a metallic shield and an outer protective jacket. The shield provides good resistance to external interference and helps control the electrical characteristics of the cable.

| Type  | Typical Use                       | Advantages                         | Limitations                                     |
|-------|-----------------------------------|------------------------------------|-------------------------------------------------|
| RG-6  | Cable TV, satellite and broadband | Good shielding and RF performance  | Bulkier than twisted pair                       |
| RG-59 | Video and CCTV applications       | Flexible and widely used for video | Higher loss than some larger coax constructions |
| RG-11 | Longer RF/video runs              | Lower loss than smaller coax types | Thicker and less flexible                       |

## 5. Fibre-Optic Cable

Fibre-optic cable transmits information using pulses of light through a glass or plastic optical medium. Unlike copper, fibre is not affected by electromagnetic interference and provides electrical isolation.

| Type                    | Basic Characteristic                    | Advantages                                     | Typical Applications                |
|-------------------------|-----------------------------------------|------------------------------------------------|-------------------------------------|
| Single-mode fibre (SMF) | Small core; one main propagation mode   | Long distance, high bandwidth, low attenuation | Telecom, backbone, long links       |
| Multimode fibre (MMF)   | Larger core; multiple propagation modes | Cost-effective for shorter optical links       | Data centres, building/campus links |

### 5.1 Advantages of Fibre

- Very high bandwidth.
- Long transmission distances.

- Immune to electromagnetic interference.
- Electrical isolation between network devices.
- Suitable for backbone and high-speed links.

### 5.2 Limitations of Fibre

- Higher installation and equipment cost.
- Termination and testing require specialised tools.
- Fibre must be handled carefully to avoid damage or contamination.

## 6. Comparison of Major Cable Types

| Parameter         | UTP/STP                    | Coaxial               | Fibre                        |
|-------------------|----------------------------|-----------------------|------------------------------|
| Signal            | Electrical                 | Electrical            | Light                        |
| Bandwidth         | High for modern categories | Good for RF/broadband | Very high                    |
| EMI immunity      | UTP: lower; STP: better    | Good                  | Excellent                    |
| Installation      | Easy                       | Moderate              | More specialised             |
| Cost              | Low to moderate            | Moderate              | Higher                       |
| Long-distance use | Limited                    | Moderate              | Excellent                    |
| Common use        | LAN access                 | TV/RF/video           | Backbone/data centre/telecom |

## 7. Straight-Through and Crossover Ethernet Cables

The conductors inside an Ethernet cable are terminated according to a wiring standard. The two standards commonly studied are T568A and T568B.

### 7.1 Straight-Through Cable

A straight-through cable uses the same wiring standard at both ends. Therefore, the cable can be terminated as T568A–T568A or T568B–T568B.

### 7.2 Crossover Cable

A traditional crossover cable uses T568A at one end and T568B at the other. The transmit and receive pairs are therefore crossed.

| Cable            | End A | End B | Traditional Example                      |
|------------------|-------|-------|------------------------------------------|
| Straight-through | T568A | T568A | Same wiring standard at both ends        |
| Straight-through | T568B | T568B | Common Ethernet patch-cable arrangement  |
| Crossover        | T568A | T568B | Traditional PC-to-PC or switch-to-switch |

#### MODERN NETWORK NOTE

Many modern Ethernet interfaces support Auto-MDI/MDI-X. Because of this, a traditional crossover cable is often not required for direct device-to-device connections. The traditional distinction is still important for this laboratory practical.

## 8. T568A and T568B Colour Codes

| Pin | T568A        | T568B        |
|-----|--------------|--------------|
| 1   | White/Green  | White/Orange |
| 2   | Green        | Orange       |
| 3   | White/Orange | White/Green  |
| 4   | Blue         | Blue         |
| 5   | White/Blue   | White/Blue   |
| 6   | Orange       | Green        |
| 7   | White/Brown  | White/Brown  |
| 8   | Brown        | Brown        |

#### MEMORY TIP

T568A begins with the green pair. T568B begins with the orange pair. The green and orange pairs are the two pairs that change positions between A and B.

## 9. Tools and Materials Required

- Cat5e or Cat6 Ethernet cable.
- Two RJ45/8P8C modular plugs.
- Crimping tool.
- Cable stripper and cutter.
- Cable tester (recommended).
- Cable boot / plug cover if required.
- T568A/T568B colour-code reference.

## 10. Procedure for Making an Ethernet Cable

1. Thread the cable boot onto the cable if a boot is being used.
2. Strip about 1–1.5 cm of the outer jacket carefully without damaging the conductors.
3. Separate and untwist the four twisted pairs only as much as necessary.
4. Choose the required wiring arrangement: T568A or T568B.
5. Arrange all eight conductors in the correct colour order.
6. Straighten the conductors and trim their ends evenly.
7. Insert the conductors completely into the RJ45 plug while maintaining the correct order.
8. Check that the outer jacket enters the plug far enough for the strain-relief section to grip it.
9. Place the plug in the crimping tool and crimp it firmly.
10. For a straight-through cable, repeat the same wiring standard at the second end.
11. For a traditional crossover cable, use the opposite standard at the second end.
12. Test the completed cable using a cable tester before using it in the network.

## 11. Cable Testing and Troubleshooting

| Tester Result                    | Meaning                          | Likely Problem                         |
|----------------------------------|----------------------------------|----------------------------------------|
| 1-1, 2-2, 3-3 ... 8-8            | Correct straight-through mapping | Cable is wired correctly               |
| Correct A-to-B crossover mapping | Correct crossover                | Cable is wired as intended             |
| Open / missing pin               | No continuity                    | Broken wire or poor crimp              |
| Two pins connected incorrectly   | Short/miswire                    | Stray conductor or incorrect insertion |
| Wrong colour order               | Miswire                          | Incorrect T568A/T568B sequence         |
| Intermittent connection          | Unreliable contact               | Poor crimp, damaged plug or cable      |

## 12. Student Observation

Record the actual observations made during the laboratory session. The cable category should be identified from the markings printed on the cable jacket.

| Observation             | Record                         |
|-------------------------|--------------------------------|
| Cable category used     | _____                          |
| Cable type              | UTP / STP / FTP / Other: _____ |
| Number of twisted pairs | _____                          |
| Connector used          | RJ45 / 8P8C                    |
| End A wiring            | T568A / T568B                  |
| End B wiring            | T568A / T568B                  |
| Cable made              | Straight-through / Crossover   |
| Cable tester result     | Pass / Fail                    |
| Network test            | Connected / Not connected      |

## 13. Answers to the Manual's Student Observation Questions

### 1. What is the difference between a cross cable and a straight cable?

A straight-through cable uses the same wiring standard at both ends, such as T568B at both ends. A traditional crossover cable uses T568A at one end and T568B at the other, crossing the relevant transmit and receive pairs.

### 2. Which type of cable is used to connect two PCs?

Traditionally, a crossover cable is used for a direct PC-to-PC Ethernet connection. However, modern network interfaces with Auto-MDI/MDI-X can often use a straight-through cable as well.

### 3. Which cable is used to connect a router/switch to your PC?

Traditionally, a straight-through Ethernet cable is used for PC-to-switch and PC-to-router connections.

### 4. Find the category of twisted-pair cable used in your lab.

Read the marking printed on the cable jacket. It may be marked CAT5E, CAT6, CAT6A or another category. The category should be recorded from the actual cable used in the laboratory.

### 5. Write your understanding, challenges faced and output received.

Sample observation: I understood how the four twisted pairs are arranged and terminated in an RJ45 plug. The main challenge was maintaining the correct colour sequence while inserting the conductors into the connector. After crimping, the cable was tested with a cable tester. The cable tester showed the expected wire mapping and the completed cable was suitable for the intended connection.

## 14. Advantages and Disadvantages – Quick Revision

| Cable   | Advantages                                 | Disadvantages                                        |
|---------|--------------------------------------------|------------------------------------------------------|
| UTP     | Low cost, flexible, easy to install        | Less protection from EMI                             |
| STP     | Better protection from interference        | More expensive; installation/grounding needs care    |
| Coaxial | Good shielding and RF performance          | Bulkier and less common for modern LAN access        |
| Fibre   | High bandwidth, long distance, EMI immune  | More expensive and requires specialised installation |
| DAC     | Low latency and economical for short links | Short reach                                          |
| AOC     | High speed and light weight                | More costly than passive copper and has fixed ends   |